

## Lesson 3-3: Determine Equivalent Ratios Using Tables

Grade 6 Mathematics · Standard 6.AT.3a

Name: \_\_\_\_\_ Date: \_\_\_\_\_

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

### Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

#### Question 1 SELECTED RESPONSE · 1 POINT

A recipe uses 3 eggs for every 5 cups of flour. How many eggs are needed for 15 cups of flour?

- ☐ A 6
- ☐ B 9
- ☐ C 10
- ☐ D 12

#### Question 2 SELECTED RESPONSE · 1 POINT

Which row does NOT belong in a ratio table for 4:5?

- ☐ A 6:10
- ☐ B 8:10
- ☐ C 12:15
- ☐ D 16:20

#### Question 3 SELECTED RESPONSE · 1 POINT

A ratio table shows 3:7, 6:14, and 9:21. What is the next row?

- ☐ A 10:24
- ☐ B 12:28
- ☐ C 12:24
- ☐ D 15:28

**Question 4**    SELECTED RESPONSE · 1 POINT

A ratio table for 2:5 starts with 2:5, 4:10. What is the third row?

- ☐ A 5:10
- ☐ B 6:15
- ☐ C 6:12
- ☐ D 8:15

**Question 5**    SELECTED RESPONSE · 1 POINT

In a ratio table, the ratio of sugar to butter is 3:2. If you use 12 cups of sugar, how many cups of butter do you need?

- ☐ A 6
- ☐ B 8
- ☐ C 9
- ☐ D 10

**Question 6**    SELECTED RESPONSE · 1 POINT

A ratio table shows 5:8, 10:16, 15:24. What scale factor was used from the first row to the third row?

- ☐ A 2
- ☐ B 3
- ☐ C 5
- ☐ D 8

**Part B — Test-Format Questions**

These match the state test's formats: two-part questions, select-all, and written responses.

**Question 7****TWO-PART QUESTION****Part A**

A ratio table pairs 3 cups of flour with 2 cups of water. If the recipe is scaled to use 12 cups of flour, how much water is needed?

*Fill in the bubble next to the one best answer.*

- ☐ A 6 cups
- ☐ B 8 cups
- ☐ C 11 cups
- ☐ D 24 cups

**Part B**

*Answer Part A before Part B — Part B asks about your reasoning.*

Why is 11 cups an incorrect answer for Part A?

- ☐ A It divides instead of multiplying.
- ☐ B It uses the wrong conversion factor.
- ☐ C It adds 9 to both terms instead of multiplying both by the same factor.
- ☐ D It reverses flour and water.

**Question 8****SELECT ALL THAT APPLY**

Select ALL statements that are true about ratio tables and equivalent ratios:

*Mark the box next to EVERY answer that is true. More than one is correct.*

- ☐ A Multiplying both terms of a ratio by 3 produces an equivalent ratio.
- ☐ B Dividing both terms by a common factor produces an equivalent ratio.
- ☐ C Adding 5 to both terms produces an equivalent ratio.
- ☐ D Every row of a correct ratio table simplifies to the same ratio.
- ☐ E A ratio table can be extended in either direction, up or down.

**Part C — Show Your Thinking**

Answer in writing, the way the state test's constructed responses work.

**Question 9**

WRITTEN RESPONSE · 2 POINTS

A recipe uses 4 cups of flour for every 3 cups of sugar. A student filled in a ratio table: 4:3, 8:6, 12:9, 14:12. Find and fix the error in the last row. Explain what went wrong.

*Write your answer in complete sentences. Show or explain your mathematical thinking.*

---

---

---

---

---

---

---

---

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.